SHARP DISCRETE HEDGING RATES WITH A PRESCRIBED CALL

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Abstract. A call can complete a stochastic-volatility market in continuous time while its changing exposure limits discrete hedging. In a Brownian stock model with bounded Volterra volatility, we determine sharp mean-square errors using one prescribed call with the target's maturity and full Brownian information. The bounds cover all square-integrable total stock/call gains on uniform and optimal deterministic dates, uniformly across a specified family from a latent-driver digital to smooth bounded targets, for each fixed power-kernel exponent $p > -1/2$. In the rough range, optimizing dates improves digital error from order $n^{-1/4}$ to $n^{-1/2}$. In that range, each fixed smooth target retains optimal error of order $(\log n)^2/n$: the completing call's terminal inefficiency persists after payoff smoothing.

1. Continuous completion and discrete execution

A European call can supply the stock market's missing Brownian exposure at every time before maturity while remaining inefficient on a finite trading grid. Its exposure may become small near maturity or vary too much within a cell. We determine how this loss interacts with target regularity when both holdings and deterministic trading dates are optimized.

As maturity approaches, less future volatility remains to affect the call's price. Its exposure to the volatility driver also changes with stock moneyness during a holding period. More frequent adjustments can reduce the hedging error, but their timing depends on the exposure of both the call and the target. We study this adjustment problem with known model parameters and both Brownian drivers observed as they evolve.

1.1. Relation to earlier work.

Payoff regularity and the allocation of trading dates are central to hedging approximation. Geiss and Gobet [1], equation (1) and Theorems 0.5.4, 0.5.6 and 0.5.8, study a selected stock-delta discretization for a stock-terminal payoff; their fractional-smoothness bounds and graded dates separate target regularity from date allocation. Gobet and Landon [6], Theorems 3.1–3.3, optimize a different criterion: trade count times error quadratic variation, almost surely, with attainment by stopping-time delta schemes under regularity and ellipticity assumptions.

Calls allow higher-order corrections to stock hedges. Gobet and Makhlouf [2], Definition 1 and Theorems 3 and 6, analyze a prescribed delta–gamma strategy using calls and exchange options expiring at $T_2 > T$. Their Itô expansions control remainders through payoff regularity and Greek ratios; Lemmas 9–10 and the mesh threshold use the positive maturity gap. Their bounds include a critical logarithmic regime, with generic tightness discussed for the uniform selected strategy below fractional regularity one. Finer rates at regularity one remain unresolved there. Brodén and Wiktorsson [3] obtain an $n^{-3/2}$ MSE expansion for a selected equidistant delta–gamma strategy with a call target and a later-expiry call hedge under local-volatility assumptions.

Pathwise results also permit common-maturity options. Armstrong and Ionescu [4], Theorem 3.1, prove convergence with bounded Greek-matching holdings and controlled rough-integral representations, allowing untraded pricing signals. Armstrong and Das [5], Theorem 1.3 and Corollary 1.4, use C^3 pricing functions, bounded derivative-cancelling holdings and suitable vanishing variation. Here the call shares the target's maturity, and its exposure degenerates

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near that date. We determine necessary and attainable MSE orders over all admitted stock/call gains and deterministic schedules, uniformly in target smoothing.

1.2. **Model, targets and admissible gains.** Fix known $T, S_0, k, c > 0$ and $p > -1/2$. Under a fixed pricing law $\mathbb{Q}$, independent standard Brownian motions W, B generate the full augmented joint natural filtration $\mathcal{F}$. Cash is identically one. Define

$$Y_t = c\int_0^t (t-r)^p\, dB_r, \qquad f(y) = 1 + \tfrac{1}{2}\tanh y, \qquad \sigma_t = f(Y_t), \qquad dS_t = S_t\sigma_t\, dW_t, \tag{1}$$

and trade, besides S, only the single prescribed call

$$C_t = \mathbb{E}_{\mathbb{Q}}[(S_T - k)_+ \mid \mathcal{F}_t], \qquad 0 \le t \le T. \tag{2}$$

The square-integrable power kernel gives a continuous version of Y: splitting translated kernels near their endpoints gives an L^2 increment modulus of a positive power (with any smaller power at a borderline), and Gaussian higher moments give continuity. The stock is the strictly positive stochastic exponential. With

$$m = \tfrac{1}{4}, \qquad M = \tfrac{9}{4}, \qquad m \le \sigma_t^2 \le M, \tag{3}$$

conditioning on B gives stock moments of every finite positive order, and S is a true square-integrable martingale. The call is also square integrable and $0 \le C_t \le S_t$. The trader observes the driver B, which is not a traded asset.

All targets pay at T. For $a \in [0,1]$ put

$$D_0 = \mathbf{1}_{\{B_T>0\}}, \qquad D_a = \Phi\left(\frac{B_T}{a\sqrt{T}}\right) \quad (a>0), \qquad U_a = T(1+a^2), \tag{4}$$

where Φ, φ are the standard normal distribution function and density. Symmetry gives $\mathbb{E}D_a = 1/2$. Smoothing changes the target, not the call maturity or trading horizon.

Definition 1.1. On a deterministic n-cell grid $\pi = (0 = t_0 < \cdots < t_n = T)$, the admissible holdings are the finite $\mathcal{F}_{t_i}$-measurable pairs (θ_i^S, θ_i^C), constant on their respective cells, whose total scalar cell gain

$$G_i = \theta_i^S(S_{t_{i+1}} - S_{t_i}) + \theta_i^C(C_{t_{i+1}} - C_{t_i})$$

belongs to $L^2(\mathbb{Q})$. Define

$$R_\pi(p,a) = \inf_\theta \mathbb{E}\left[\left(D_a - \tfrac{1}{2} - \sum_i G_i\right)^2\right], \tag{5}$$

$$R_n^{\mathrm{det}}(p,a) = \inf_{\pi:\, n \text{ deterministic cells}} R_\pi(p,a), \qquad R_n^{\mathrm{unif}}(p,a) = R_{(iT/n)_{i=0}^n}(p,a). \tag{6}$$

No maximum-mesh premise is imposed on the infimum. In particular, separate global legwise energy is not required when two large legs cancel in the total gain. These quantities are mean-square errors (MSE); their square roots are L^2 norms. All comparison constants below may depend on the fixed parameters p, T, S_0, k, c, and not on a, the grid, or n, unless explicitly indicated.

The target energy will be needed both for necessity and date allocation.

Lemma 1.2. *For $t < T$,*

$$\mathbb{E}_t D_a = \Phi\left(\frac{B_t}{\sqrt{U_a - t}}\right), \qquad D_a - \tfrac{1}{2} = \int_0^T y_t^{(a)}\, dB_t, \qquad y_t^{(a)} = \frac{\varphi(B_t/\sqrt{U_a - t})}{\sqrt{U_a - t}}. \tag{7}$$

The coefficient $y^{(a)}$ is a martingale between strict preterminal dates. Its energy density is increasing and satisfies

$$F_a(t) := \mathbb{E}[(y_t^{(a)})^2] = \frac{1}{2\pi\sqrt{U_a^2 - t^2}}, \qquad \mathrm{Var}(D_a) = \frac{1}{2\pi}\arcsin(T/U_a), \tag{8}$$

$$TF_a(T(1-r)) = \frac{1}{2\pi\sqrt{(a^2+r)(2+a^2-r)}} \asymp (a^2+r)^{-1/2}, \quad 0 < r \le 1, \tag{9}$$

$$\frac{F_a'(t)}{F_a(t)} = \frac{t}{U_a^2 - t^2} \le \frac{1}{T-t}. \tag{10}$$

The comparison in (9) *is uniform in $a \in [0,1]$.*

Proof. Gaussian convolution of Φ proves the conditional price, including the digital at $a = 0$. Its space derivative is $y^{(a)}$, and the backward heat equation removes the drift in Itô's formula. Differentiating the heat equation, or convolving its normal density, also proves $\mathbb{E}_s y_u^{(a)} = y_s^{(a)}$ for $s \le u < T$. A one-dimensional Gaussian integral gives

$$\mathbb{E}e^{-B_t^2/(U_a - t)} = \sqrt{\frac{U_a - t}{U_a + t}},$$

which gives the formula for F_a. Integrating F_a from 0 to T gives $\mathrm{Var}(D_a)$. The conditional martingale converges in L^2 to D_a, so the representation extends to T, also at $a = 0$, without assigning a square-integrable terminal value to $y^{(0)}$. The remaining assertions follow by substitution and differentiation; $1 \le 2 + a^2 - r \le 3$. □

2. Sharp rates and the smoothing crossover

The allocation of dates balances the target's remaining Brownian energy against the call's local hedging inefficiency. In residual time, the density below combines both quantities; its quantiles place more dates where either makes a long holding period costly. Put

$$\begin{aligned} &q = \max(2, 2p+2), \qquad \varepsilon_n = n^{-1/(q-1)}, \qquad \delta_n = n^{-1/q}, \\ &f_{q,a}(r) = r^{-q/2}(a^2+r)^{-1/4}, \qquad \Lambda_{n,q}(a) = \int_{\varepsilon_n}^1 f_{q,a}(r)\,dr. \end{aligned} \tag{11}$$

The allocation density $f_{q,a}$ is distinct from the volatility map f.

Theorem 2.1. *For every fixed $p > -1/2$ and fixed positive T, S_0, k, c, there exist $0 < c_* \le C_* < \infty$ and n_0, independent of $a \in [0,1]$, such that for all $n \ge n_0$ and all $a \in [0,1]$,*

$$R_n^{\mathrm{det}}(p,a) \asymp \frac{\Lambda_{n,q}(a)^2}{n}, \qquad R_n^{\mathrm{unif}}(p,a) \asymp \frac{\delta_n}{\sqrt{a^2+\delta_n}}. \tag{12}$$

For $q > 2$ the first expression is comparable to

$$\frac{\varepsilon_n}{\sqrt{a^2+\varepsilon_n}}. \tag{13}$$

For $q = 2$ it is comparable to

$$\begin{cases} n^{-1/2}, & na^2 \le 1, \\ (1+\log(na^2))^2/(na), & na^2 > 1. \end{cases} \tag{14}$$

An attaining sequence of deterministic grids is given by

$$\int_{r_j}^1 f_{q,a}(r)\,dr = \frac{j}{n-1}\Lambda_{n,q}(a), \quad 0 \le j < n, \qquad t_j = T(1-r_j), \quad t_n = T, \tag{15}$$

where $r_0 = 1 > \cdots > r_{n-1} = \varepsilon_n$. The stock/call projection holdings defined in (27) *attain the upper order on these dates.*

Here $\asymp$ denotes two-sided orders, not a leading-constant limit. The constants cover arbitrary deterministic sequences $a = a_n$. The crossover is $a \asymp n^{-1/(2q)}$ on uniform dates and $a \asymp n^{-1/(2(q-1))}$ on optimized dates. No uniformity as $p \downarrow -1/2$, $H \downarrow 0$, or $c \downarrow 0$ is asserted.

Corollary 2.2. *For the digital target,*

$$R_n^{\mathrm{unif}}(p,0)=\Theta(n^{-1/(2q)}),\qquad R_n^{\mathrm{det}}(p,0)=\Theta(n^{-1/(2(q-1))}).$$

In the rough family $p=H-1/2$, fixed $0<H<1/2$, and the smooth model $p=1$, these become

Digital experiment	*Uniform-date MSE*	*Deterministic-optimal MSE*
Rough, fixed $H\in(0,1/2)$	$\Theta(n^{-1/4})$	$\Theta(n^{-1/2})$
Smooth, $Y_t=c\int_0^t B_r\,dr$	$\Theta(n^{-1/8})$	$\Theta(n^{-1/6})$

For every fixed $a>0$, in the rough or $p=0$ market,

$$R_n^{\mathrm{det}}(p,a)=\Theta((\log n)^2/n),\qquad R_n^{\mathrm{unif}}(p,a)=\Theta(n^{-1/2}). \tag{16}$$

For fixed $a>0$ and $p>0$, the corresponding orders are $n^{-1/(q-1)}$ and $n^{-1/q}$.

The smooth model identity follows by stochastic Fubini. Norm exponents are half of the MSE exponents: the digital table gives $(1/8,1/4)$ in the rough case and $(1/16,1/12)$ in the smooth case. Optimizing dates improves the digital rates in both models. Smoothing the target improves them further, but leaves the squared logarithm in (16). That penalty is therefore not confined to a discontinuous payoff: it is necessary over every admitted gain and every deterministic n-cell schedule.

Two features of the call determine the exponent q. Independent stock noise changes the call's missing-driver exposure through moneyness, producing the scale $h/(T-s)^2$. For $p>0$, decay of that exposure relative to a stock-orthogonal quadratic component gives the stronger scale $h/(T-s)^{2p+2}$. This explains the slower rates in the specified smooth-volatility model: the loss comes from the behavior of the hedging instrument's exposure near maturity.

The call's exact projection on each trading cell gives upper bounds that control both exposure variation and stock-delta leakage. We then use the cell bounds to allocate trading dates: a counted refinement gives a lower bound for every deterministic schedule, and equal-mass spacing attains the same order. At $q=2$, a smooth target leaves the allocation integral $\int dr/r$, which accounts for the squared logarithm.

3. The call and its interval projection

Write

$$\begin{aligned} A_{uv}&=\int_u^v\sigma_r^2\,dr,\qquad L_t=\log(S_t/k),\qquad h_0(y)=2f(y)f'(y),\\ V(L,A)&=\frac{\varphi(L/\sqrt{A}+\sqrt{A}/2)}{2\sqrt{A}},\qquad \Gamma_t=c\int_t^T(u-t)^p h_0(Y_u)\,du. \end{aligned} \tag{17}$$

The quantities A_{tT} and Γ_t are future functionals, not observed state coordinates. Direct differentiation gives

$$0<h_0\le\tfrac{3}{2},\qquad h_0(y)\ge\tfrac{1}{2}e^{-2|y|},\qquad |h_0'|\le 3h_0,\qquad M-f(y)^2\le Ce^{-2y}. \tag{18}$$

For spot $x>0$ and total variance $A>0$, let

$$\mathcal{B}(x,A)=x\Phi(d_+)-k\Phi(d_-),\qquad d_\pm=\frac{\log(x/k)}{\sqrt{A}}\pm\frac{\sqrt{A}}{2}.$$

Then $\mathcal{B}_x=\Phi(d_+)$, $\mathcal{B}_A=xV$, and $\mathcal{B}_A=\frac{1}{2}x^2\mathcal{B}_{xx}$.

Proposition 3.1. *The call admits the predictable representation*

$$\begin{aligned} C_t&=\mathbb{E}_t\mathcal{B}(S_t,A_{tT}),\qquad dC_t=S_t\sigma_t\delta_t\,dW_t+b_t\,dB_t,\\ \delta_t&=\mathbb{E}_t\Phi(d_+(S_t,A_{tT})),\qquad \rho_t=\mathbb{E}_t[V(L_t,A_{tT})\Gamma_t],\qquad b_t=S_t\rho_t>0. \end{aligned} \tag{19}$$

Positivity holds $dt \otimes \mathbb{Q}$-almost everywhere before T. The coefficients are square integrable. Moreover $U_t = \mathbb{E}_t[S_T \mathbf{1}_{\{S_T>k\}}] = S_t\delta_t$ is a square-integrable martingale with

$$\begin{aligned} dU_t &= S_t\sigma_t(\delta_t + 2\mathbb{E}_t V)\, dW_t + S_t\delta_B(t)\, dB_t, \\ \delta_B(t) &= \mathbb{E}_t[V\Gamma_t(\tfrac{1}{2} - L_t/A_{tT})], \qquad |S_t\delta_B(t)| \le b_t\left(\tfrac{1}{2} + \frac{|L_t|}{m(T-t)}\right). \end{aligned} \tag{20}$$

In these formulas, V is evaluated at (L_t, A_{tT}).

Proof. Conditional on $\mathcal{F}_t$ and the whole B path, the future stock log-return is normal with variance $A = A_{tT}$, proving the price formula. Appendix A justifies Gaussian differentiation and approximation at the strike for every $p > -1/2$. To compute the exposures, let

$$J_t = \int_t^T f'(Y_u)c(u-t)^p\, dW_u, \qquad D_t^0 = \int_t^T \sigma_u f'(Y_u)c(u-t)^p\, du = \Gamma_t/2.$$

The Gaussian derivative of S_T in direction B_t is $S_T(J_t - D_t^0)$. Given the B path, the pair consisting of the future stock log integral and J_t has covariance D_t^0. Gaussian regression and integration at the strike threshold give

$$\mathbb{E}[\mathbf{1}_{\{S_T>k\}} S_T(J_t - D_t^0) \mid \mathcal{F}_t, B] = S_t D_t^0 \frac{\varphi(d_+(S_t, A))}{\sqrt{A}} = S_t V(L_t, A)\Gamma_t.$$

The positive term on the right arises from integration at the strike threshold. The W derivative gives $S_t\sigma_t\Phi(d_+)$. Predictable projection identifies (19). Since the integrand of Γ_t is strictly positive and $A \ge m(T-t) > 0$, conditional expectation preserves positivity. Also

$$b_t \le CS_t(T-t)^{p+1/2}, \qquad 0 < \delta_t < 1,$$

so stock moments give product-space integrability. The closed derivative and martingale arguments ensure the predictable versions of these formulas. Finally, differentiating the positive-variance Black convolution for $x\Phi(d_+)$ gives (20); its compact-preterminal approximation justification is also in Appendix A. □

The traded exposure determinant is $S_t\sigma_t b_t > 0$. Thus for any $X \in L^2(\mathcal{F}_T)$ with Brownian coefficients (x_t, y_t), continuous replication uses

$$\theta_t^C = y_t/b_t, \qquad \theta_t^S = \frac{x_t - S_t\sigma_t\delta_t y_t/b_t}{S_t\sigma_t}. \tag{21}$$

Replication is defined in the completed multivariate total-exposure norm

$$\mathbb{E}\int_0^T \left[(\theta_t^S S_t\sigma_t + \theta_t^C S_t\sigma_t\delta_t)^2 + (\theta_t^C b_t)^2\right] dt = \mathbb{E}\int_0^T (x_t^2 + y_t^2)dt < \infty. \tag{22}$$

The integral is constructed by localizing both legs on the same bounded holding norms and stopping the total price quadratic variation. Each localized leg is square integrable; their sum converges in total-exposure norm and hence in L^2 martingale path norm. The same truncation multiplies both legs, preserving cancellation. No inverse-exposure floor or unweighted holding moment follows. For D_a the energy is $\mathrm{Var}(D_a)$. This continuous convention is distinct from Definition 1.1; continuous completion alone gives no quantitative grid rate.

3.1. **Closed cell gain spaces.** Fix a cell $[s,t]$, $h = t-s$, and write $\mathbb{E}_s = \mathbb{E}[\,\cdot \mid \mathcal{F}_s]$. Define

$$\beta_s = \frac{\mathbb{E}_s[\Delta S\Delta C]}{\mathbb{E}_s[(\Delta S)^2]}, \qquad v_s = \mathbb{E}_s[(\Delta C - \beta_s\Delta S)^2], \qquad g_s = \mathbb{E}_s\int_s^t b_u\, du, \qquad \zeta_{s,h} = 1 - \frac{g_s^2}{hv_s}. \tag{23}$$

Conditional Itô isometry gives

$$v_s = \mathbb{E}_s\int_s^t b_u^2\, du + \mathbb{E}_s\int_s^t S_u^2\sigma_u^2(\delta_u - \beta_s)^2\, du > 0. \tag{24}$$

The second term is the stock-delta leakage from a single constant stock/call ratio. The conditional stock variance is positive, $0 < \beta_s < 1$, and the two-asset conditional Gram determinant is $\mathbb{E}_s[(\Delta S)^2]v_s > 0$. Cauchy–Schwarz proves $0 \le \zeta \le 1$, and more explicitly

$$\zeta = \frac{\mathbb{E}_s \int_s^t (b_u - g_s/h)^2 du + \mathbb{E}_s \int_s^t S_u^2 \sigma_u^2 (\delta_u - \beta_s)^2 du}{v_s}.$$

Both types of variation enter the same bounded inefficiency.

Proposition 3.2. *For any L^2 target measurable with respect to B, write $X = \mathbb{E}X + \int_0^T y_u \, dB_u$ with B-predictable y. Put*

$$A_s^X = \mathbb{E}_s \int_s^t y_u^2 du, \qquad c_s^X = \mathbb{E}_s \int_s^t y_u b_u \, du.$$

The exact conditional optimal residual on this cell is

$$\mathfrak{r}_{s,t}(X) = A_s^X - \frac{(c_s^X)^2}{v_s}. \tag{25}$$

It is attained by call units c_s^X/v_s and stock units $-\beta_s c_s^X/v_s$, with total conditional gain energy at most A_s^X. For a deterministic grid, with initial cash $\mathbb{E}X$,

$$R_\pi(X) = \sum_i \mathbb{E}\mathfrak{r}_{t_i,t_{i+1}}(X). \tag{26}$$

The infimum covers exactly all the gains in Definition 1.1.

Proof. Let G_s be the conditional Gram matrix of $(\Delta S, \Delta C)$. It is finite positive definite almost surely. For any finite $\mathcal{F}_s$-measurable vector θ, localization on $\{\|\theta\| \le j\}$ proves, as an extended nonnegative identity,

$$\mathbb{E}_s[(\theta^S \Delta S + \theta^C \Delta C)^2] = \theta^\top \mathsf{G}_s \theta.$$

Consequently the weighted space with squared norm $\mathbb{E}\theta^\top \mathsf{G}_s \theta$ is complete: multiplication by $\mathsf{G}_s^{1/2}$ identifies it with $L^2(\mathcal{F}_s; \mathbb{R}^2)$, and the pointwise inverse is measurable and finite. Its scalar gain map is an isometry, so its image is closed. For admissible holdings the conditional mean of the total gain is zero, by the same localization and L^2 convergence. Distinct cell spaces are therefore orthogonal.

The target increment $\int_s^t y \, dB$ has zero conditional stock covariance, whereas its covariance with the stock-orthogonalized call is c_s^X. Conditional positivity of the joint target/asset Gram matrix (first for rational vectors, then by continuity) gives $(c_s^X)^2/v_s \le A_s^X$. Two-dimensional orthogonalization gives the stated holdings and residual. The energy bound makes these holdings admissible. Localization and Cauchy–Schwarz justify cross moments with every other admissible total gain, so the resulting gains are orthogonal projections onto the closed cell gain spaces. Earlier and later target increments are orthogonal to this cell's gain. Summing the cell projections proves (26). □

3.2. **A projection strategy and its freezing error.** If y is a martingale between strict preterminal dates, projection of the frozen increment $y_s \Delta B$ uses

$$\theta_s^C = \frac{y_s g_s}{v_s}, \qquad \theta_s^S = -\beta_s \frac{y_s g_s}{v_s}, \qquad \mathbb{E}_s[(\theta_s^S \Delta S + \theta_s^C \Delta C)^2] = \frac{y_s^2 g_s^2}{v_s} \le h y_s^2. \tag{27}$$

For $F(u) = \mathbb{E}y_u^2$ define

$$J_\pi = \sum_i h_i \mathbb{E}[y_{t_i}^2 \zeta_{t_i,h_i}], \qquad E_B(\pi) = \sum_i \int_{t_i}^{t_{i+1}} (F(u) - F(t_i)) \, du. \tag{28}$$

The distance squared of $\sum_i y_{t_i} \Delta_i B$ to the grid gain space is J_π. Martingality of y and Itô isometry give $\| \int y \, dB - \sum_i y_{t_i} \Delta_i B \|_2^2 = E_B(\pi)$, with the integrable endpoint understood by limits. The strategy (27) thus has MSE at most

$$(\sqrt{J_\pi} + \sqrt{E_B(\pi)})^2 \le 2(J_\pi + E_B(\pi)). \tag{29}$$

For the lower bound, we retain the time-varying target coefficient inside (25). This preserves the lower order uniformly in target smoothing.

4. Lower bounds over all admissible gains

4.1. **A lower bound from independent stock noise.** The first obstruction comes from independent stock noise. It does not rely on the upper estimates, and it applies to any L^2 B-measurable target, including a path-dependent one.

Theorem 4.1. *Fix $\ell > 0$, put $\tau = T - s$ and $\kappa = m\ell^2/(4M^2)$. On $\{L_s \geq \ell\}$, if $\tau \leq \ell/M$, then for every deterministic cell $[s,t] \subseteq [s,T]$,*

$$\mathfrak{r}_{s,t}(X) \geq \mathbb{E}_s \int_s^t y_u^2 \frac{\kappa(u-s)}{\tau^2 + \kappa(u-s)}\, du. \tag{30}$$

This is a lower bound for the true target and the entire admitted gain class.

Proof. Let $\mathcal{G}_u = \mathcal{F}_s \vee \sigma(B_r - B_s : s \leq r \leq u)$, completed. Given $\mathcal{G}_u$, the stock innovation is normal:

$$S_u = S_s e^{\sqrt{A_{su}} Z - A_{su}/2}, \qquad Z \sim N(0,1), \qquad m(u-s) \leq A_{su} \leq M(u-s).$$

Future B is independent of Z. The exact Black convolution identity

$$\mathbb{E}_Z\Big[S_s e^{\sqrt{a_*} Z - a_*/2} V(L_s + \sqrt{a_*} Z - a_*/2, A)\Big] = S_s V(L_s, a_* + A) \tag{31}$$

follows by differentiating the Black semigroup with respect to its positive variance. Consequently the W-averaged missing-driver exposure is

$$\begin{aligned} \mu_u &:= \mathbb{E}[b_u \mid \mathcal{G}_u] = S_s \mathbb{E}[V(L_s, A_{sT}) \Gamma_u \mid \mathcal{G}_u] > 0, \\ \frac{\partial_L \mu_u}{\mu_u} &= \tfrac{1}{2} - L_s R_u, \qquad R_u = \frac{\mathbb{E}[V\Gamma_u / A_{sT} \mid \mathcal{G}_u]}{\mathbb{E}[V\Gamma_u \mid \mathcal{G}_u]} \in [(M\tau)^{-1}, (m\tau)^{-1}]. \end{aligned} \tag{32}$$

The L derivative varies initial log spot with B history fixed, and includes the spot factor S_s; hence $\partial_L \log(SV) = 1/2 - L/A$. On strict preterminal intervals the positive variance bounds and bounded Γ_u justify differentiation and Gaussian integration by parts. Projection onto the constant and the first normal coordinate of independent stock noise gives

$$\begin{aligned} \mathbb{E}[b_u Z \mid \mathcal{G}_u] &= \sqrt{A_{su}}\, \partial_L \mu_u, \\ \mathbb{E}[b_u^2 \mid \mathcal{G}_u] &\geq \mu_u^2 + A_{su} (\partial_L \mu_u)^2 \geq w_s(u) \mu_u^2, \qquad w_s(u) = 1 + \kappa(u-s)/\tau^2. \end{aligned}$$

Indeed $L_s \geq \ell$ and $\tau \leq \ell/M$ imply $|1/2 - L_s R_u| \geq \ell/(2M\tau)$.

Since y_u is B-predictable it is $\mathcal{G}_u$-measurable, so the exact covariance is $c_s^X = \mathbb{E}_s \int_s^t y_u \mu_u \, du$. Weighted conditional Cauchy–Schwarz on time and future B yields

$$(c_s^X)^2 \leq \left(\mathbb{E}_s \int_s^t \frac{y_u^2}{w_s(u)}\, du\right) \left(\mathbb{E}_s \int_s^t w_s(u) \mu_u^2 \, du\right) \leq v_s \mathbb{E}_s \int_s^t \frac{y_u^2}{w_s(u)}\, du. \tag{33}$$

The last step uses (24). Subtracting this captured energy from A_s^X proves (30). The weighted inequality retains the time-varying coefficient y_u. For cells ending at T, monotone convergence extends the nonnegative energy bounds from strict preterminal times. The target and call coefficient L^2 energies control convergence of the covariance. Localization from Proposition 3.2 covers every finite admitted holding. □

Averaging over stock noise gives a uniform unconditional bound. Conditional on the full B path, for $s \geq T/2$, L_s is normal with mean $L_0 - A_{0s}/2$ and variance $A_{0s} \in [mT/2, MT]$. Thus

$$\mathbb{Q}(L_s \geq \ell \mid \mathcal{F}_T^B) \geq p_\ell > 0, \qquad p_\ell = 1 - \Phi\left(\frac{\ell + |L_0| + MT/2}{\sqrt{mT/2}}\right). \tag{34}$$

In particular $\mathbb{E}[y_u^2\mathbf{1}_{\{L_s\ge\ell\}}]\ge p_\ell\mathbb{E}y_u^2$. The choice $\ell=MT$ makes the maturity restriction hold throughout. If $F(u)=\mathbb{E}y_u^2$ is increasing, the integral Chebyshev inequality for the two increasing functions F and $1-1/w_s$ gives

$$\mathbb{E}\mathfrak{r}_{s,t}(X)\ge p_\ell W_{s,t}\,\mathcal{F}(\kappa h/\tau^2),\qquad W_{s,t}=\int_s^t F(u)\,du,\qquad \mathcal{F}(x)=1-\frac{\log(1+x)}{x}. \tag{35}$$

The inequality $(F(u)-F(v))(\psi(u)-\psi(v))\ge 0$ holds pointwise. Integrating over the square with bounded truncations of F, followed by monotone convergence, also covers integrable endpoint singularities. Finally

$$\mathcal{F}(x)=\int_0^1\frac{xv}{1+xv}\,dv\ge\frac{x}{2(1+x)}\ge\tfrac14\min(1,x). \tag{36}$$

The bound controls the whole target energy in a terminal cell. The positive probability in (34) is independent of target smoothing.

4.2. **A stock-orthogonal Hermite obstruction.** For nonnegative powers the exposure-decay mechanism is stronger. The witness below combines the first and second Gaussian chaoses to remove the stock component. Its exact conditional form gives a bound for every cell length.

Proposition 4.2. *For $p\ge0$, set $q_0=2p+2$, $G=3c/[2(p+1)]$ and $\kappa_0=4m^2/[G^2(2+MT)]$. For every L^2 B-measurable target and every cell,*

$$\mathfrak{r}_{s,t}(X)\ge A_s^X\frac{\kappa_0 h}{\tau^{q_0}+\kappa_0 h}. \tag{37}$$

Proof. With $\mathcal{G}_u$ as above define the B-martingales

$$P_u=\mathbb{E}[\mathcal{B}(S_s,A_{sT})\mid\mathcal{G}_u],\qquad H_u=\mathbb{E}[S_s^2\mathcal{B}_{xx}(S_s,A_{sT})\mid\mathcal{G}_u].$$

They are square integrable, since $0\le\mathcal{B}\le S_s$ and $0<H_u\le S_s/\sqrt{2\pi m\tau}$. Also $\mathbb{E}[C_t\mid\mathcal{G}_t]=P_t$ and $P_s=C_s$. Given $\mathcal{G}_t$, put $a_*=A_{st}$ and take the standard normal stock innovation Z. The witness

$$\Xi=Z^2-1-\sqrt{a_*}Z$$

satisfies

$$\mathbb{E}[\Xi\mid\mathcal{G}_t]=0,\quad \mathbb{E}[\Xi^2\mid\mathcal{G}_t]=2+a_*,\quad \mathbb{E}[\Delta S\,\Xi\mid\mathcal{G}_t]=0,\quad \mathbb{E}[C_t\Xi\mid\mathcal{G}_t]=a_*H_t. \tag{38}$$

For completeness, if $P_aF(x)=\mathbb{E}F(xe^{\sqrt{a}Z-a/2})$, differentiation of the Gaussian convolution density gives

$$\mathbb{E}[F(xe^{\sqrt{a}Z-a/2})Z]=\sqrt{a}\,\partial_{\log x}P_aF,\quad \mathbb{E}[F(xe^{\sqrt{a}Z-a/2})(Z^2-1)]=a\,\partial^2_{\log x}P_aF.$$

Their combination is $ax^2\partial_x^2P_aF$. Applying this identity to the conditional call function at t and adding the variances gives the last identity in (38). For $F(x)=x$ it gives zero. Gaussian moments prove the other identities. At $t=T$, the convolution has variance $a_*\ge mh>0$, so the same calculation applies to the kinked payoff of linear growth.

Conditional variance decomposition and Cauchy–Schwarz against Ξ give, for any finite $\mathcal{F}_s$-measurable stock coefficient,

$$v_s\ge V_{\mathrm{sig}}+\frac{m^2h^2}{2+MT}\mathbb{E}_sH_t^2,\qquad V_{\mathrm{sig}}=\mathbb{E}_s(P_t-P_s)^2. \tag{39}$$

The Gaussian derivative rule in Appendix A, conditional on $\mathcal{F}_s$, gives

$$dP_u=\nu_u\,dB_u,\qquad \nu_u=\mathbb{E}[\mathcal{B}_A(S_s,A_{sT})\Gamma_u\mid\mathcal{G}_u]\le\tfrac12G\tau^{p+1}H_u.$$

Here $\Gamma_u\le G(T-u)^{p+1}\le G\tau^{p+1}$ and $\mathcal{B}_A=\frac12S_s^2\mathcal{B}_{xx}$. Martingale increase of $\mathbb{E}_sH_u^2$ implies

$$V_{\mathrm{sig}}\le\frac{G^2\tau^{q_0}h}{4}\mathbb{E}_sH_t^2,\qquad v_s\ge(1+\kappa_0h/\tau^{q_0})V_{\mathrm{sig}}. \tag{40}$$

The true target increment is $\mathcal{G}_t$-measurable; its call covariance is its covariance with P_t-P_s, and its stock covariance is zero. Thus $(c_s^X)^2\le A_s^XV_{\mathrm{sig}}$. Substitution in (25), together with (40),

proves the result, also when $V_{\mathrm{sig}} = 0$. Localization extends the conditional comparisons to all admissible holdings. □

Moneyness gives exponent 2 for $p < 0$; exposure decay gives $2p + 2$ for $p > 0$. Either works at $p = 0$. This explains $q = \max(2, 2p + 2)$ independently of the upper analysis.

5. Upper bounds and target weighting

To match the lower bounds, we must control hedging error even when the call's exposure is small or the observed history is extreme. We bound its variation relative to its current exposure, allowing that exposure to approach zero. The appendices establish the following bound for each decision history.

For $\tau = T - s$ define the predictor

$$\mu_s(v) = c \int_0^s (s + v - r)^p \, dB_r, \qquad 0 \le v \le \tau.$$

For $p \ge 0$ put

$$M_s = \frac{p+1}{\tau^{p+1}} \int_0^\tau v^p |\mu_s(v)| \, dv, \qquad \mathcal{A}_s = C \exp\{C(M_s + |L_s|)\}. \tag{41}$$

For $-1/2 < p < 0$, put $H = p + 1/2$, $\alpha = p + 1 \in (1/2, 1)$ and

$$K_s = \sup_{0 \le v \le \tau} |\mu_s(v)|, \qquad \mathcal{K}_s = C e^{CK_s} (1 + K_s)^b (1 + |L_s|)^b, \tag{42}$$

where $b < \infty$ is a fixed exponent that can be enlarged in a finite number of estimates.

Lemma 5.1. *For $h \le \tau/2$ the call's inefficiency satisfies, on all states,*

$$\zeta_{s,h} \le \begin{cases} \min(1, C\mathcal{A}_s^2 h \tau^{-(2p+2)}), & p \ge 0, \\ \min(1, C\mathcal{K}_s^2 h \tau^{-2}), & -1/2 < p < 0. \end{cases} \tag{43}$$

Proof. Appendix B proves the nonnegative branch: Black-weight entropy gives relative sensitivity, conditional Gaussian derivatives and positive-mixture moments give relative exposure variation, and the delta martingale controls stock leakage. Appendix C proves the rough branch. Its extra steps are refined innovation covariance, separate entropy estimates for V and $V\Gamma_s$, a positive finite-energy Cameron–Martin shift, and absorption of the random prefix predictor before Gaussian Poincaré. Both proofs conclude with a localized conditional comparison, then use $\zeta \le 1$ on ineligible states. The global strategy uses the holdings in (27), so this comparison imposes no inverse-exposure integrability condition. □

To obtain bounds uniform in target smoothing, define a probability on past decision information only by

$$\left.\frac{d\mathbb{Q}_s^a}{d\mathbb{Q}}\right|_{\mathcal{F}_s} = \frac{(y_s^{(a)})^2}{F_a(s)}. \tag{44}$$

Its past B covariance is

$$\mathrm{Cov}_{\mathbb{Q}_s^a}(B_u, B_v) = \min(u, v) - \frac{2uv}{U_a + s}, \qquad u, v \le s. \tag{45}$$

Indeed the weight is proportional to $\exp(-B_s^2/(U_a - s))$. Multiplication of the $N(0, s)$ density changes its variance to $s(U_a - s)/(U_a + s)$, and its conditional Brownian bridge remains unchanged. Past W stays independent Brownian motion. Every conditional future option expectation in ζ is still under $\mathbb{Q}$.

Lemma 5.2. *All fixed finite moments of the envelopes in* (41)–(42) *are bounded uniformly under $\mathbb{Q}_s^a$, for $a \in [0, 1]$, $s < T$. Consequently*

$$\mathbb{E}_{\mathbb{Q}_s^a} \zeta_{s,h} \le C \min(1, h(T - s)^{-q}), \qquad h \le (T - s)/2. \tag{46}$$

Proof. The covariance in (45) subtracts a nonnegative rank-one form from Brownian covariance. Hence the variance of every centered linear Gaussian functional of past B is no greater than its original variance. For $p \geq 0$, each $\mu_s(v)$ has variance at most $c^2T^{2p+1}/(2p+1)$. Jensen first for the normalized v^p weight and then for the convex function $e^{\eta x^2}$ gives a uniformly bounded small exponential square moment for M_s.

For $p < 0$, choose $\gamma_0 \in (-p, 1/2)$ and let $\widehat{B}_x = B_s - B_{s-x}$ be the reversed increments. Integration by parts gives

$$\int_0^s (v+x)^p \, d\widehat{B}_x = (v+s)^p \widehat{B}_s - p \int_0^s (v+x)^{p-1} \widehat{B}_x \, dx. \tag{47}$$

At $v = 0$ the lower boundary vanishes because $p + \gamma_0 > 0$. The absolute value, uniformly in $v \geq 0$, is at most $C[\widehat{B}]_{\gamma_0;[0,s]} s^{p+\gamma_0} = Cs^H [X]_{\gamma_0;[0,1]}$, where $X_x = s^{-1/2}\widehat{B}_{sx}$. Under the tilted law its increments are centered normal with variance at most their time length. Thus for $x \geq 1$,

$$\mathbb{Q}_s^a \left(\sup_{j \geq 0} 2^{j\gamma_0} \max_{k < 2^j} |X_{(k+1)2^{-j}} - X_{k2^{-j}}| > x \right) \leq \sum_{j \geq 0} 2^{j+1} e^{-x^2 2^{j(1-2\gamma_0)}/2} \leq Ce^{-c_0 x^2}. \tag{48}$$

Deterministic dyadic chaining controls the Hölder seminorm by a fixed multiple of this supremum. Consequently K_s has a uniform small exponential square moment. This argument does not need independent increments under the tilted law.

Conditional on past B, L_s is normal with mean $L_0 - A_{0s}/2$ and variance $A_{0s} \leq MT$, also under $\mathbb{Q}_s^a$. Its absolute exponential moments of any fixed order are uniformly bounded. Hölder's inequality combines these with the predictor square moments to bound every finite envelope moment. Averaging (43) and also using $\zeta \leq 1$ proves (46). □

5.1. **A deterministic upper functional and a terminal lower functional.** For any finite deterministic grid put

$$\begin{gathered} r_i = (T - t_i)/T, \qquad d_i = (t_{i+1} - t_i)/T, \qquad W_{a,i} = \int_{t_i}^{t_{i+1}} F_a(u) \, du, \\ \mathcal{C}_{q,a}(\pi) = \sum_i W_{a,i} \min(1, d_i/r_i^q). \end{gathered} \tag{49}$$

Let $\mathcal{C}_{q,a}^{\mathrm{tail}}$ be the same sum restricted to cells starting at or after $T/2$.

Proposition 5.3. *Uniformly over $a \in [0,1]$ and all finite deterministic grids,*

$$R_\pi(p,a) \leq C\mathcal{C}_{q,a}(\pi), \qquad R_\pi(p,a) \geq c\mathcal{C}_{q,a}^{\mathrm{tail}}(\pi). \tag{50}$$

The frozen-target projection strategy attains the upper estimate, including its freezing error. No early-cell two-sided equivalence is asserted.

Proof. The lower estimate follows from (35)–(36) when $p < 0$ and Proposition 4.2 when $p \geq 0$, together with the exact residual sum. Fixed powers of T are absorbed into constants.

For the upper estimate consider first a cell with $h \leq \tau/2$. Equations (44) and (46) bound its J contribution by $ChF_a(s)\min(1, d/r^q) \leq CW_a \min(1, d/r^q)$. On such a cell (10) gives comparable values of F_a and

$$E_{B,i} = \int_s^{s+h} (F_a(u) - F_a(s)) du \leq Ch^2 F_a(s)/\tau \leq CW_a d/r.$$

Also $E_{B,i} \leq W_a$. Since $r \leq 1$ and $q \geq 2$, $d/r \leq d/r^q$, so freezing is absorbed by the same bounded penalty. If $h > \tau/2$, then $d/r^q \geq r^{1-q}/2 \geq 1/2$ while $J_i \leq hF_a(s) \leq W_a$ and $E_{B,i} \leq W_a$. Summing over the cells and applying (29) proves the upper estimate. □

6. The deterministic date budget

6.1. **Refinement and necessity for arbitrary schedules.** We first check that refinement respects the exact scalar total-gain convention.

Lemma 6.1. *If π' is a deterministic refinement of π, then $R_\pi(X) \geq R_{\pi'}(X)$.*

Proof. Let $\mathbf{P} = (S, C)$, and on a coarse cell let $G = \theta \cdot (\mathbf{P}_t - \mathbf{P}_s) \in L^2$ with finite $\mathcal{F}_s$-measurable θ. For deterministic $u \in [s, t]$,

$$\theta \cdot (\mathbf{P}_u - \mathbf{P}_s) = \mathbb{E}[G \mid \mathcal{F}_u]. \tag{51}$$

On the event $\{\|\theta\| \le j\} \in \mathcal{F}_s$, the coefficients are bounded, so the L^2 martingale property proves the identity. This localization uses only the total-gain moment assumption. The truncated terminal total gains converge in L^2; so do their conditional expectations. Their almost sure limit is the finite-coefficient left side. Each refined gain is therefore a difference of two L^2 conditional expectations. The coarse holdings are therefore admissible on the refined grid and preserve the total terminal gain. Taking infima proves the assertion. □

Fix an arbitrary deterministic n-cell grid, $q \ge 2$, $a \in [0, 1]$, and $\varepsilon = \varepsilon_n \le 1/4$. We refine the grid by adding $T/2$, $T(1-\varepsilon)$ and all dates with dyadic residual fractions between ε and $1/2$. At most $O_q(\log n)$ dates are added. The band $[\varepsilon, 1/2]$ has at most $N \le C_q n$ cells, each satisfying

$$r_i/2 \le r_{i+1} < r_i, \qquad d_i = r_i - r_{i+1}. \tag{52}$$

Lemma 6.1, the terminal lower functional, and (9) give

$$R_\pi(p, a) \ge c \sum_{i \text{ in band}} \frac{d_i^2}{\sqrt{a^2 + r_i}(r_i^q + d_i)}. \tag{53}$$

Indeed $W_{a,i} \ge c d_i/\sqrt{a^2 + r_i}$ and $\min(1, x) \ge x/(1 + x)$. Set $v_i = d_i f_{q,a}(r_i)$. Weighted Cauchy–Schwarz gives

$$\sum_i \frac{d_i^2}{\sqrt{a^2 + r_i}(r_i^q + d_i)} = \sum_i \frac{v_i^2}{1 + d_i/r_i^q} \ge \frac{(\sum_i v_i)^2}{N + \sum_i d_i/r_i^q}. \tag{54}$$

The denominator is at most

$$C_q n + \int_\varepsilon^{1/2} r^{-q} dr \le C_q' n, \qquad \varepsilon^{1-q} = n. \tag{55}$$

On a cell with comparable residuals, the decreasing density satisfies $f_{q,a}(r_{i+1})/f_{q,a}(r_i) \le 2^{q/2+1/4}$. Thus

$$\sum_i v_i \ge c_q \int_\varepsilon^{1/2} f_{q,a}(r) dr \ge c_q' \Lambda_{n,q}(a). \tag{56}$$

The omitted $[1/2, 1]$ integral is uniformly bounded in a, and $[1/4, 1/2]$ supplies a uniform positive lower mass. Combining (53)–(56) proves

$$R_\pi(p, a) \ge c \Lambda_{n,q}(a)^2/n \tag{57}$$

for every original deterministic n-cell schedule. The refinement retains the entire critical logarithmic integral while adding only $O_q(\log n)$ dates, so the denominator remains of order n.

6.2. **Uniform evaluation of the allocation integral.** The following elementary estimate will also control the last cell in the attaining construction.

Lemma 6.2. *For fixed $q \ge 2$, $\varepsilon = n^{-1/(q-1)}$ and all sufficiently large n, uniformly in $a \in [0, 1]$, Λ^2/n has the orders in* (13)–(14)*. In all cases*

$$\frac{\varepsilon}{\sqrt{a^2 + \varepsilon}} \le C_q \Lambda^2/n. \tag{58}$$

Proof. If $a^2 \le \varepsilon$, then $a^2 + r \asymp r$ on $[\varepsilon, 1]$. With $\beta = q/2 + 1/4 > 1$,

$$\Lambda \asymp \int_\varepsilon^1 r^{-\beta} dr \asymp \varepsilon^{1-\beta}, \qquad \Lambda^2/n \asymp \varepsilon^{1/2}.$$

If $a^2 > \varepsilon$ and $q > 2$, the portion $[\varepsilon, a^2]$ is at most $C_q a^{-1/2} \varepsilon^{1-q/2}$, while the portion above a^2 is at most $C_q a^{3/2-q}$, no larger in order. A matching lower bound follows by integrating on $[\varepsilon, 2\varepsilon]$, where $a^2 + r \le 3a^2$. This works even near transition; $2\varepsilon \le 1$. Hence

$$\Lambda \asymp a^{-1/2} \varepsilon^{1-q/2}, \qquad \Lambda^2/n \asymp \varepsilon/a.$$

Together these prove (13) uniformly, including a near 1.

If $q = 2$, then $\varepsilon = 1/n$. For $a^2 > \varepsilon$, splitting at a^2 gives a lower portion comparable to $a^{-1/2}\log(a^2/\varepsilon)$ and an upper portion comparable to $a^{-1/2}-1$. Their sum is uniformly comparable to

$$a^{-1/2}\{1 + \log(a^2/\varepsilon)\}. \tag{59}$$

To see the additive 1 explicitly, when $a^2/\varepsilon \geq e$ it is supplied by the logarithmic portion, including a near 1. When $1 < a^2/\varepsilon < e$, n large makes a^2 small, and the upper portion supplies a fixed multiple of $a^{-1/2}$. Squaring proves (14). The final-cell estimate follows from the digital regime when $a^2 \leq \varepsilon$, from ε/a when $q > 2$, and from the additional factor $(1 + \log(a^2/\varepsilon))^2 \geq 1$ when $q = 2$. □

6.3. **Quantile dates and an attaining strategy.** For the quantile dates in (15), write $\lambda_n = \Lambda_{n,q}(a)/(n-1)$. Since $a, r \leq 1$ and $q \geq 2$,

$$f_{q,a}(r) \geq 2^{-1/4}/r, \qquad \Lambda_{n,q}(a) \leq \int_\varepsilon^1 r^{-q/2-1/4}dr.$$

The latter gives $\lambda_n \leq C_q n^{-(2q-1)/(4(q-1))} \to 0$ uniformly in a. Equal mass in each preterminal cell therefore implies

$$\log(r_i/r_{i+1}) \leq 2^{1/4}\lambda_n \longrightarrow 0. \tag{60}$$

For all large n, independently of a, every preterminal cell has $d_i \leq r_i/2$. Since the density is decreasing,

$$d_i f_{q,a}(r_i) \leq \int_{r_{i+1}}^{r_i} f_{q,a}(r)dr = \lambda_n.$$

Comparable residuals and (9) then give

$$W_{a,i}\min(1, d_i/r_i^q) \leq C\frac{d_i^2}{r_i^q\sqrt{a^2+r_i}} = C[d_i f_{q,a}(r_i)]^2 \leq C\lambda_n^2. \tag{61}$$

The remaining target energy on the final cell, of length $T\varepsilon$, is at most

$$C\int_0^\varepsilon (a^2+r)^{-1/2}dr = \frac{2C\varepsilon}{\sqrt{a^2+\varepsilon}+a} \asymp \frac{\varepsilon}{\sqrt{a^2+\varepsilon}} \leq C_q\Lambda^2/n. \tag{62}$$

Summing (61) over the preterminal cells and adding (62) bounds the upper functional. Proposition 5.3, including the freezing error, then gives an MSE of at most $C\Lambda^2/n$.

The units are precisely (27) with $y = y^{(a)}$. They are finite and decision-measurable. Their conditional total cell energy is at most $h_i(y^{(a)}_{t_i})^2$, and its expected sum is at most

$$\sum_i h_i F_a(t_i) \leq \int_0^T F_a(u)du = \mathrm{Var}(D_a) \leq \tfrac{1}{4}.$$

The constructed total gains are therefore admissible. These grids attain the sharp asymptotic order.

6.4. **Uniform dates.** For $d = 1/n$ and residual fractions $r_j = j/n$, the upper functional is, up to a uniform constant,

$$\sum_{j=1}^n \left[\int_{(j-1)/n}^{j/n} (a^2+r)^{-1/2}dr\right]\min(1, d/r_j^q). \tag{63}$$

Cells below $\delta = d^{1/q}$, together with the rounding cell, contribute at most $C\delta/\sqrt{a^2+\delta}$. Earlier cells are bounded by

$$Cd\int_\delta^1 r^{-q}(a^2+r)^{-1/2}dr \leq \frac{Cd\delta^{1-q}}{(q-1)\sqrt{a^2+\delta}} = C_q\frac{\delta}{\sqrt{a^2+\delta}}.$$

The last $\lfloor n\delta \rfloor$ cells give a matching lower bound. For large n they start after $T/2$ and satisfy $d/r_j^q \geq 1$. The terminal lower functional captures their whole target energy, at least

$$c \int_0^{\lfloor n\delta \rfloor / n} (a^2 + r)^{-1/2} dr \geq c' \frac{\delta}{\sqrt{a^2 + \delta}},$$

since $\lfloor n\delta \rfloor / n \asymp \delta$. This proves the second order in (12); the upper is realized by the same admitted projections on uniform dates.

Completion of the proof of Theorem 2.1. The all-date lower bound (57) and the quantile construction prove the deterministic assertion. The uniform-date argument proves its counterpart. Lemma 6.2 evaluates the deterministic expression. The two crossover scales follow by comparing a^2 with δ_n and ε_n, respectively. All thresholds and constants used above are independent of a. □

6.5. **Digital schedules and the persistent smooth-target logarithm.** At $a = 0$, integration of $r^{-\beta}$, $\beta = q/2 + 1/4$, in (15) gives

$$r_j = \left[1 + \frac{j}{n-1}(\varepsilon_n^{1-\beta} - 1)\right]^{-1/(\beta-1)}. \tag{64}$$

Thus the rough model admits

$$t_j = T\left[1 - \left\{1 + \frac{j}{n-1}(n^{1/4} - 1)\right\}^{-4}\right], \quad j < n, \qquad t_n = T,$$

with last cell T/n. The smooth model admits

$$t_j = T\left[1 - \left\{1 + \frac{j}{n-1}(n^{5/12} - 1)\right\}^{-4/5}\right], \quad j < n, \qquad t_n = T,$$

with last cell $Tn^{-1/3}$. These sequences attain the stated asymptotic orders. Substituting $a = 0$ and fixed $a > 0$ in the theorem proves Corollary 2.2.

For $q = 2$ and fixed $a > 0$, the target energy is bounded at maturity, while $f_{2,a}(r) \asymp a^{-1/2} r^{-1}$ near zero. Integrating this density gives the logarithm, and the lower bound shows that it persists under every deterministic schedule. For $p > 0$ the stronger exposure-decay obstruction produces the powers stated in the corollary.

7. Conclusion

The prescribed call's exposure governs the best discrete hedging rate in the frictionless model studied here, where parameters are known and both Brownian drivers are observed. Variation between adjustments and decay near maturity determine the error incurred over a holding period. Balancing this local inefficiency against the target's Brownian energy gives explicit deterministic schedules that attain the optimal asymptotic order over all admitted stock/call holdings.

Smoothing the target improves the rates, while the call's terminal inefficiency persists. For every fixed smooth target in the specified family, the rough-volatility model has optimal mean-square error of order $(\log n)^2/n$. The specified smooth-volatility example converges more slowly because its call exposure decays more severely. Discrete hedging difficulty therefore depends jointly on target regularity and the behavior of the security used to supply the missing exposure.

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Appendix A. Gaussian differentiation and payoff approximation

A.1. **Closed derivative, duality, and conditional inequalities.** Let $\mathsf{H} = L^2(0,T;\mathbb{R}^2)$, in coordinate order (W,B), and $I(h) = \int h^W dW + \int h^B dB$. On smooth finite Gaussian cylinders define

$$DF = \sum_j \partial_j F_0(I(h_1),\dots,I(h_d))h_j.$$

Finite-dimensional normal integration by parts gives, for deterministic h and smooth cylindrical G,

$$\mathbb{E}[G\langle DF,h\rangle] = \mathbb{E}[F\{GI(h) - \langle DG,h\rangle\}]. \tag{65}$$

Compact cutoffs extend this to cylinders with the necessary Gaussian square moments. If $F_j \to 0$ in L^2 and $DF_j \to Z$ in $L^2(\mathsf{H})$, (65) gives $\mathbb{E}[G\langle Z,h\rangle] = 0$. Bounded smooth cylinders are dense: finite Brownian increment fields exhaust $\mathcal{F}_T$, and smooth functions are dense for a finite normal law. A countable dense set of h forces $Z = 0$. Thus D is closable; denote its closed domain by $\mathbb{D}^{1,2}$.

For $F \in \mathbb{D}^{1,2}$ and C^1 ψ with bounded continuous derivative, closure proves $D\psi(F) = \psi'(F)DF$. Indeed values converge by Lipschitz continuity, and for derivatives one separates $DF_j - DF$ from $(\psi'(F_j) - \psi'(F))DF$, using bounded domination. This also proves the rule after smoothing ψ with locally uniform derivative limits.

Choose a bounded smooth past cylinder $G \in \mathcal{F}_s$ and $h = \mathbf{1}_{(s,u]}e_j$. Since $\langle DG,h\rangle = 0$, duality gives

$$\mathbb{E}[FG(W_u^j - W_s^j)] = \mathbb{E}\left[G\int_s^u D_r^j F\,dr\right].$$

Density in the past field and in predictable step processes, with Itô isometry and Brownian martingale representation, identify the Brownian coefficient of $F - \mathbb{E}F$ with the predictable projection $\Pi_{\mathrm{pred}}DF$. This argument also holds conditional on a fixed decision history for fresh Brownian increments. Conditional differentiation commutes with integration over independent future coordinates. The identity holds for cylinders and extends by derivative closure and contraction of conditional expectation.

For reference the Gaussian inequalities used below are

$$\operatorname{Var} X \le \mathbb{E}\|DX\|^2, \qquad \operatorname{Ent}(X^2) \le 2\mathbb{E}\|DX\|^2, \qquad \operatorname{Ent}(F) = \mathbb{E}[F\log(F/\mathbb{E}F)]. \tag{66}$$

One derivation uses the Ornstein–Uhlenbeck semigroup P_v. For cylinders, Gaussian integration by parts yields

$$\operatorname{Var} X = 2\int_0^\infty \mathbb{E}\|DP_vX\|^2 dv, \quad \operatorname{Ent}(F) = \int_0^\infty \mathbb{E}\frac{\|DP_vF\|^2}{P_vF}\,dv.$$

Gradient commutation $DP_v = e^{-v}P_vD$, Jensen, and weighted Cauchy–Schwarz bound the first by $\mathbb{E}\|DX\|^2$ and the second by $\frac{1}{2}\mathbb{E}\|DF\|^2/F$. Taking $F = X^2$ gives the second inequality. The calculation holds first for positive bounded F (or $F+\eta$); truncation and derivative-domain approximation give the general case. Finite-coordinate exhaustion proves the Brownian versions and their conditional versions.

We also use the entropy variational inequality

$$\mathbb{E}_\nu\psi \le H(\nu \mid \mathbb{Q}) + \log\mathbb{E}e^\psi, \qquad \log\mathbb{E}F \ge \mathbb{E}_\nu\log F - H(\nu\mid\mathbb{Q}), \quad F>0. \tag{67}$$

Both follow by Jensen under ν applied to $e^\psi/(d\nu/d\mathbb{Q})$ or $F/(d\nu/d\mathbb{Q})$; truncation handles integrable limits. In particular for a centered Gaussian Z of variance v,

$$|\mathbb{E}_\nu Z| \le \sqrt{2vH(\nu\mid\mathbb{Q})}, \qquad \mathbb{E}_\nu|Z| \le \sqrt{2v(H(\nu\mid\mathbb{Q}) + \log 2)}. \tag{68}$$

These bounds follow by optimizing the parameter in $\psi = \lambda Z$ or $\lambda|Z|$, using $\mathbb{E}e^{\lambda|Z|} \le 2e^{\lambda^2 v/2}$.

A.2. **Volterra stock and the call kink.** For $K(u,r) = c(u-r)^p \mathbf{1}_{\{r<u\}}$,

$$\int_0^T \int_r^T K(u,r)^2 du\, dr = \frac{c^2 T^{2p+2}}{(2p+1)(2p+2)} < \infty. \tag{69}$$

The row $k_u = K(u,\cdot)$ is continuous in $L^2(0,T)$ and has bounded row norms; cutting out the endpoints and then using dominated convergence proves this also for negative p. On a dyadic partition, let $\eta_j(u)$ be the left endpoint and let P_j project onto deterministic step kernels. Set $k_u^j = P_j k_{\eta_j(u)}$ and $Y_u^j = I_B(k_u^j)$. The support is before $\eta_j(u)$, so $f(Y^j)$ is an adapted step process. Row continuity gives $k^j \to k$ in $L^2(du\,dr)$ and $Y^j \to Y$ in $L^2(du \otimes \mathbb{Q})$.

The log stock variable

$$\mathsf{L}_j = \int_0^T f(Y_u^j) dW_u - \tfrac{1}{2} \int_0^T f(Y_u^j)^2 du$$

is a smooth finite-coordinate function. Finite differentiation, Itô isometry, and bounded continuous f' and ff' show that it and its derivatives converge to

$$\mathsf{L} = \log(S_T/S_0), \qquad D_r^W \mathsf{L} = \sigma_r, \qquad D_r^B \mathsf{L} = J_r - D_r^0, \tag{70}$$

where

$$J_r = \int_r^T f'(Y_u) K(u,r)\, dW_u, \qquad D_r^0 = \int_r^T \sigma_u f'(Y_u) K(u,r) du.$$

For example, the derivative integrands converge by splitting the kernel error from the bounded coefficient error times k_u, followed by dominated convergence. Hilbert-valued Itô isometry in r proves convergence of J. Only the squared kernel in (69) is needed, not its fourth power.

For the exponential, we condition on the whole B path and write $A_* = \int_0^T \sigma_u^2 du$ and $Q_r = \int_r^T f'(Y_u)^2 K(u,r)^2 du$. The Gaussian pair $(\int_0^T \sigma dW, J_r)$ has covariance D_r^0. Completing squares under the exponential tilt gives

$$\mathbb{E}[S_T^2 (J_r - D_r^0)^2 \mid B] = S_0^2 e^{A_*}(Q_r + (D_r^0)^2) \le S_0^2 e^{MT}(1+MT) Q_r. \tag{71}$$

Integrating in r proves product-space square integrability. The W derivative is controlled by bounded σ. A smooth upper cutoff of the exponential with derivative between 0 and e^x converges to e^x; dominated convergence using (71) and stock moments then proves

$$D_r^W S_T = S_T \sigma_r, \qquad D_r^B S_T = S_T (J_r - D_r^0)$$

in the closed domain. The conditional normal law has positive variance, so $\mathbb{Q}(S_T = k) = 0$. Smooth approximations of $(x-k)_+$ with slopes in $[0,1]$ and vanishing value error therefore give

$$D_r^W (S_T - k)_+ = \mathbf{1}_{\{S_T > k\}} S_T \sigma_r, \qquad D_r^B (S_T - k)_+ = \mathbf{1}_{\{S_T > k\}} S_T (J_r - D_r^0). \tag{72}$$

Thus the call payoff belongs to $\mathbb{D}^{1,2}$ with the stated derivatives.

Integration over future W identifies the exposure. Given $\mathcal{F}_t, B$, let $N = \int_t^T \sigma dW$, with variance $A = A_{tT}$ and covariance D_t^0 with J_t. Gaussian regression gives $J_t = (D_t^0/A)N + R$ with independent centered R. With $a_0 = \log(k/S_t) + A/2$,

$$\mathbb{E}[e^{N - A/2} \mathbf{1}_{\{N > a_0\}}] = \Phi(d_+), \quad \mathbb{E}[N e^{N - A/2} \mathbf{1}_{\{N > a_0\}}] = A\Phi(d_+) + \sqrt{A}\varphi(d_+).$$

The second identity includes the boundary term obtained by differentiating the normal density. Substitution in (72) gives the calculation in Proposition 3.1. All integrations against predictable tests are justified by the proved product L^2 bounds. Since $\Gamma_t \le C(T-t)^{p+1}$ and $V \le C(T-t)^{-1/2}$, the conditional exposures in (19) belong to product L^2.

Joint measurability follows from the Volterra integrals. On compact preterminal intervals, their unprojected future-functional formulas are L^2-continuous, using continuity of S, Y, L^1 translation continuity of the integrable power kernel, and the square-integrable stock maximum. Left-step conditional approximations thus converge to their predictable projections and identify these with $\mathbb{E}_t$ versions for almost every time. Strict positivity of the unprojected $S_t V \Gamma_t$ is preserved by predictable projection: on a predictable positive-measure set where the projection vanished its positive integral would have to be zero. This proves the asserted $dt \otimes \mathbb{Q}$ positivity without a deterministic floor.

To derive (20), we approximate the discontinuous terminal variable $S_T\mathbf{1}_{\{S_T>k\}}$ by $H_j(S_T)$, where $H_j(x) = xq_j(x)$ and q_j are smooth threshold approximations in $[0,1]$. For each j, H_j has bounded derivative and the preceding closure applies. It converges in L^2 to the terminal variable. Conditional future-W integration gives its coefficients by the derivatives of $\mathbb{E}_Z H_j(xe^{\sqrt{A}Z-A/2})$. On $A \geq m\eta > 0$ the first spot and variance derivatives can be taken on the Gaussian density. They converge to the derivatives of $x\Phi(d_+)$, dominated by $C_\eta x$ in these normalized derivatives because $|H_j(xe^{\sqrt{A}Z-A/2})| \leq xe^{\sqrt{A}Z-A/2}$ and Gaussian score moments are bounded. Stock moments yield coefficient convergence on $[0, T-\eta]$. Martingale representation and Itô isometry also give convergence of the global coefficients from terminal L^2 convergence. Uniqueness identifies them on every such compact interval. Finally

$$x\partial_x(x\Phi(d_+)) = x(\Phi(d_+) + 2V), \qquad \partial_A(x\Phi(d_+)) = xV(\tfrac{1}{2} - L/A).$$

Exhausting the preterminal intervals proves (20). The same derivative arguments apply to the bounded-variance conditional functionals used in the lower and upper proofs, with the decision history fixed.

Appendix B. Relative estimates for nonnegative powers

We first prove the stock-mixture algebra shared by both kernel branches. All calculations below fix the decision history at s, put $\tau = T - s$, and restrict $u \in [s, s+h]$ with $h \leq \tau/2$. Conditional expectations of fresh increments remain under $\mathbb{Q}$.

B.1. **Fixed-function martingales and positive Black mixtures.** Define

$$\begin{aligned} P_u &= S_u\mathbb{E}_u V(L_u, A_{uT}), \qquad \mathsf{F}_u = S_u\mathbb{E}_u[V(L_u, A_{uT})\Gamma_s], \\ \mathsf{N}_u &= \mathbb{E}[V(L_s, A_{sT}) \mid \mathcal{G}_u], \qquad \mathsf{M}_u = \mathbb{E}[V(L_s, A_{sT})\Gamma_s \mid \mathcal{G}_u]. \end{aligned} \tag{73}$$

Here the future functional Γ_s retains its index s as u varies. By (31) and the tower property, P and F are positive square-integrable martingales on this cell. Indeed $A_{uT} \geq m\tau/2$, $\Gamma_s \leq C\tau^{p+1}$, and conditional stock moments suffice. Moreover

$$\mathbb{E}[P_u \mid \mathcal{G}_u] = S_s\mathsf{N}_u, \qquad \mathbb{E}[\mathsf{F}_u \mid \mathcal{G}_u] = S_s\mathsf{M}_u, \qquad P_s = S_s\mathsf{N}_s, \quad \mathsf{F}_s = b_s = S_s\mathsf{M}_s. \tag{74}$$

Given $\mathcal{G}_u$, let $a_* = A_{su} \leq Mh$ and $L_u = L_s - a_*/2 + \sqrt{a_*}Z$. For each future constituent variance $A = A_{uT} \in [m\tau/2, M\tau]$ put

$$\begin{aligned} g_A(Z) &= S_s e^{\sqrt{a_*}Z - a_*/2} V(L_s - a_*/2 + \sqrt{a_*}Z, A), \quad R_A = g_A/\mathbb{E}_Z g_A, \\ Q_A &= a_*/A, \qquad z_A = \sqrt{a_*}\left[\tfrac{1}{2} - (L_s - a_*/2)/A\right]. \end{aligned} \tag{75}$$

Apart from its constant factor, $g_A = \exp(-Q_A Z^2/2 + z_A Z)$. Completing the Gaussian square proves the exact identity

$$\mathbb{E}_Z R_A^2 = \frac{1+Q_A}{\sqrt{1+2Q_A}} \exp\left(\frac{z_A^2}{(1+Q_A)(1+2Q_A)}\right) \leq e^{\chi}, \qquad \chi = Ch(1+L_s^2)/\tau^2. \tag{76}$$

Indeed $Q_A \leq Ch/\tau$, the logarithm of the prefactor is at most CQ_A^2, and $z_A^2 \leq Ch(1+L_s^2)/\tau^2$. Under the normalized squared weight, Z has mean $2z_A/(1+2Q_A)$ and variance $1/(1+2Q_A)$. Equivalently L_u has mean $(L_s + a_*/2)/(1+2Q_A)$ and variance $a_*/(1+2Q_A)$. Hence

$$\mathbb{E}_Z[(1+L_u^2)R_A^2] \leq C(1+L_s^2)e^{\chi}. \tag{77}$$

These estimates extend to every positive mixture with nonnegative B-functional weights, notably 1, Γ_s and Γ_u. After division by its positive W mean, the mixture is a probability mixture of the R_A, with weights proportional to the original weight times $\mathbb{E}_Z g_A$. Jensen bounds its square, also after multiplication by $1 + L_u^2$.

B.2. **Relative sensitivity for nonnegative powers.** Write $Y_{s+v} = \mu_s(v) + Z_v$, where $Z_v = c\int_0^v (v-r)^p d\widetilde{B}_r$ and $\widetilde{B}$ is fresh. Let $\mathbb{E}_V$ denote expectation under the future B law weighted by the density $V(L_s, A_{sT})/\mathbb{E}_s V$. Since $A_{sT} \in [m\tau, M\tau]$,

$$\operatorname{osc} \log V \le C(1 + L_s^2/\tau + \tau).$$

The relative entropy of this density is no greater than that oscillation. By (68),

$$\mathbb{E}_V|Z_v| \le Cv^{p+1/2}\sqrt{1 + L_s^2/\tau + \tau} \le C(1 + |L_s|), \qquad v \le \tau. \tag{78}$$

The last step uses $p \ge 0$ and $\tau \le T$. The lower bound for h_0 in (18) and Jensen, first over the product of the normalized v^p weight and the tilted law, give

$$x_s := \mathbb{E}_V \Gamma_s \ge \frac{c\tau^{p+1}}{2(p+1)} e^{-2M_s - C(1+|L_s|)}.$$

Thus, with the envelope in (41) chosen large enough,

$$r_s := \frac{\mathbb{E}_s V}{\rho_s} = x_s^{-1} \le \mathcal{A}_s \tau^{-(p+1)}, \qquad 1 + L_s^4 \le \mathcal{A}_s^2. \tag{79}$$

This ratio bound allows b_s to approach zero.

B.3. **Conditional derivatives and absorption.** For fresh derivative time $z \in [s, u]$, Appendix A gives

$$D_z A_{sT} = \Gamma_z \le C\tau^{p+1}, \qquad |D_z \Gamma_s| \le 3c\tau^p \Gamma_s. \tag{80}$$

The second inequality follows by differentiation under the integral:

$$D_z \Gamma_s = c^2 \int_z^T (r-s)^p (r-z)^p h_0'(Y_r) dr.$$

Since $p \ge 0$, we have $(r-z)^p \le \tau^p$. Combining this with $|h_0'| \le 3h_0$ and enlarging the remaining positive integral to $[s, T]$ gives the bound.

Since

$$\partial_A \log V = -\frac{1}{2A} + \frac{L_s^2}{2A^2} - \frac{1}{8}, \tag{81}$$

conditional derivative commutation gives

$$|D_z \mathsf{N}_u| \le a_s \mathsf{N}_u, \qquad |D_z \mathsf{M}_u| \le a_s \mathsf{M}_u, \qquad a_s = C(1 + L_s^2)\tau^{p-1}. \tag{82}$$

Fixed L_s, τ and the positive variance bound keep the derivatives integrable; finite Gaussian approximation and closure justify these inequalities in the derivative domain. Conditional Poincaré now gives, for either $X = \mathsf{N}_u$ or M_u, $\operatorname{Var}_s X \le h a_s^2 \mathbb{E}_s X^2$. If $h a_s^2 \le 1/2$, the variance term can be absorbed, giving

$$\mathbb{E}_s X^2 \le 2(\mathbb{E}_s X)^2, \qquad \operatorname{Var}_s X \le 2h a_s^2 (\mathbb{E}_s X)^2. \tag{83}$$

The means are $\mathsf{N}_s, \mathsf{M}_s$ by martingality.

Combining (74), (76)–(77) and (83), whenever $h a_s^2$ and χ are small, yields

$$\begin{gathered} \mathbb{E}_s[(\mathsf{F}_u/b_s - 1)^2] \le C(h a_s^2 + \chi), \qquad \mathbb{E}_s P_u^2/b_s^2 \le C r_s^2, \\ \mathbb{E}_s[(1 + L_u^2)\mathsf{F}_u^2]/b_s^2 \le C(1 + L_s^2). \end{gathered} \tag{84}$$

For example the second moment ratio of F is bounded by $e^{\chi}/(1 - h a_s^2)$, while its mean is b_s, so subtracting one gives its relative variance. The weighted assertion follows by conditional Jensen for the positive mixture and the M second moment.

B.4. **Moving exposure and stock-delta leakage.** Call a decision state eligible when

$$h(1+L_s^4)\tau^{-q} \le \eta, \qquad q = 2p+2, \tag{85}$$

where $\eta > 0$ is a sufficiently small fixed constant. This ensures the preceding smallness conditions: ha_s^2 is bounded by the left side times $C\tau^{4p}$, and χ by it times $C\tau^{2p}$. On eligible states, also $ha_s^2 + \chi \le C(1+L_s^4)h\tau^{-2}$.

For $p \ge 0$ positivity and kernel order give

$$0 \le \Gamma_s - \Gamma_u \le C(u-s)\tau^p. \tag{86}$$

Indeed replace h_0 by $3/2$ in the nonnegative kernel difference. Its integral is $[\tau^{p+1} - (\tau - (u-s))^{p+1}]/(p+1) \le (u-s)\tau^p$. Thus $0 < b_u \le \mathsf{F}_u$ and $|\mathsf{F}_u - b_u| \le Ch\tau^p P_u$. Equations (79) and (84) imply

$$\begin{aligned} \mathbb{E}_s[(b_u/b_s - 1)^2] &\le C\{(1+L_s^4)h\tau^{-2} + h^2\tau^{2p}r_s^2\} \le C\mathcal{A}_s^2 h\tau^{-q}, \\ \mathbb{E}_s[(1+L_u^2)b_u^2]/b_s^2 &\le C(1+L_s^2). \end{aligned} \tag{87}$$

The bounded factors in these power comparisons use $p \ge 0$, $\tau \le T$, $h \le \tau/2$.

The stock-delta error can be expressed using U from (20). Let $Z_u^\delta = U_u - \delta_s S_u = S_u(\delta_u - \delta_s)$. It is a square-integrable martingale starting at zero, and

$$dZ_u^\delta = \sigma_u(Z_u^\delta + 2P_u)dW_u + S_u\delta_B(u)dB_u. \tag{88}$$

Itô isometry, (20), and $T - u \ge \tau/2$ give

$$\mathbb{E}_s|Z_u^\delta|^2 \le C\int_s^u \mathbb{E}_s|Z_v^\delta|^2 dv + Cb_s^2 h\{r_s^2 + (1+L_s^2)\tau^{-2}\}.$$

Gronwall and (79) therefore imply

$$\mathbb{E}_s|Z_u^\delta|^2/b_s^2 \le C\mathcal{A}_s^2 h\tau^{-q}. \tag{89}$$

The portfolio with call units $1/b_s$ and stock units $-\delta_s/b_s$ gives a conditional comparison with ΔB. Its squared residual is

$$\mathbb{E}_s\int_s^{s+h}\left[(b_u/b_s - 1)^2 + \sigma_u^2(Z_u^\delta/b_s)^2\right]du. \tag{90}$$

The minimum conditional residual is $h\zeta_{s,h}$, so (87) and (89) bound ζ by $C\mathcal{A}_s^2 h\tau^{-q}$ on eligible states. This conditional comparison is justified by localization on bounded decision coefficients. The global strategy is the projection in (27), for which only total-gain integrability is required. On ineligible states, $\zeta \le 1$ and $\mathcal{A}_s^2 \ge 1 + L_s^4$ give the bound after increasing the fixed constant by $1/\eta$. This proves the nonnegative part of Lemma 5.1 on all states. Uniform envelope moments were proved in Lemma 5.2 without any assumption on reciprocal exposure.

Appendix C. Relative estimates for rough powers

Fix $-1/2 < p < 0$, $H = p + 1/2 \in (0, 1/2)$ and $\alpha = p + 1 \in (1/2, 1)$. We use the martingales (73) and the positive-mixture estimates of Appendix B. Throughout this appendix the initial history at s is fixed, $\tau = T - s$, and $L = L_s$. Define

$$\mathscr{D} = 1 + \tau^{-1} + L^2\tau^{-2}, \qquad \sqrt{\mathscr{D}} \le C_T\sqrt{1+L^2}/\tau. \tag{91}$$

On $A \in [m\tau, M\tau]$, (81) gives $|(\log V)_A| \le C\mathscr{D}$.

The rough kernel has two useful deterministic derivative and translation bounds:

$$D_z A_{sT} = \Gamma_z, \qquad |D_z\Gamma_s| \le C\int_z^T (r-s)^{\alpha-1}(r-z)^{\alpha-1}dr \le C\tau^{2H}, \qquad z \ge s, \tag{92}$$

$$|\Gamma_z - \Gamma_w| \le C|z-w|^\alpha, \qquad z, w \ge s. \tag{93}$$

The derivative bound in (92) follows from boundedness of h_0' and $(r-s)^{\alpha-1} \le (r-z)^{\alpha-1}$; the squared singularity is integrable since $2\alpha - 1 = 2H > 0$. For (93), suppose $z < w$. The difference separates into the integral over $[z, w]$ and the integral of the nonnegative kernel difference

$(r-w)^{\alpha-1}-(r-z)^{\alpha-1}$ on $[w,T]$. Each is bounded by a constant times $(w-z)^{\alpha}$. No time derivative of the moving exposure is assumed.

C.1. **Fresh predictors and a Gaussian multiplier.** The past predictor supremum K_s is finite and its weighted square-exponential bound was proved in Lemma 5.2. For a fresh prefix $s \le u \le s+h$ put

$$A^{\mathrm{pred}}_{s,u} = \sup_{r\in[u,T]} \left| c\int_s^u (r-v)^{\alpha-1} dB_v \right|.$$

The predictor decomposition gives $K_u \le K_s + A^{\mathrm{pred}}_{s,u}$: the old predictor is evaluated at residual argument at least $u-s$, still in the interval over which K_s takes its supremum. For $\gamma_0 \in (1-\alpha, 1/2)$, reversed-increment integration by parts (47) on the independent segment of length $u-s$ bounds its supremum by $C(u-s)^H$ times the Hölder seminorm of a scaled Brownian path. The dyadic estimate (48) (now for ordinary fresh Brownian increments) therefore proves

$$\mathbb{E}_s \exp\left(\frac{(A^{\mathrm{pred}}_{s,u})^2}{Ch^{2H}}\right) \le C. \tag{94}$$

The same bound with $u-s$ in place of h holds when $u > s$; the zero-length case is immediate.

Lemma C.1. *For a functional X in the Gaussian derivative space of the fresh prefix,*

$$\mathbb{E}_s[(A^{\mathrm{pred}}_{s,u})^2 X^2] \le Ch^{2H}\left(\mathbb{E}_s X^2 + \mathbb{E}_s \|DX\|^2_{L^2[s,u]}\right). \tag{95}$$

Proof. For nonnegative F, Jensen under $F/\mathbb{E}_s F$, or (67), gives

$$\lambda \mathbb{E}_s[(A^{\mathrm{pred}}_{s,u})^2 F] \le \mathrm{Ent}_s(F) + \mathbb{E}_s F \log \mathbb{E}_s e^{\lambda (A^{\mathrm{pred}}_{s,u})^2}.$$

With $F = X^2$ and $\lambda = (Ch^{2H})^{-1}$, the moment bound (94) and conditional logarithmic Sobolev inequality (66) give the result. Truncation and derivative-domain approximation prove the assertion for all such X. No derivative of the supremum is needed. □

This multiplier controls the contribution of fresh increments to the random predictor K_u in the derivative estimate.

C.2. **Refined innovation covariance and relative sensitivity.**

Lemma C.2. *With the envelope* (42) *enlarged if necessary,*

$$\frac{\mathbb{E}_s V(L, A_{sT})}{\mathbb{E}_s[V(L, A_{sT})\Gamma_s]} \le Ce^{2K_s}\tau^{-\alpha} + Ce^{K_s}\sqrt{\mathscr{D}} \le \mathcal{K}_s/\tau. \tag{96}$$

Proof. Write $Y_{s+r} = \mu_s(r) + Z_r$, with $Z_r = c\int_0^r (r-v)^{\alpha-1} d\beta_v$ for fresh Brownian β. Its nonnegative covariance satisfies the refined bound

$$0 \le \mathrm{Cov}(Z_r, Z_u) \le C \min(r,u)^{\alpha} \max(r,u)^{\alpha-1} \le Cr^{\alpha}u^{\alpha-1}. \tag{97}$$

For $r \le u/2$, the far kernel is bounded by $Cu^{\alpha-1}$, while the integral of the near kernel is at most Cr^{α}. For comparable r, u, Gaussian Cauchy–Schwarz and $\mathbb{E}Z_r^2 = c^2 r^{2H}/(2H)$ give the same bound. Symmetry proves the first upper bound; when $u < r$ the ratio of that bound to the last expression is u/r, and when $r \le u$ the expressions coincide.

Let $\mathbb{E}_V$ be expectation under conditional future density $V/\mathbb{E}_s V$, and set $x = \mathbb{E}_V \Gamma_s > 0$. To control the tilted innovation, we use the bounded scalar Stein solution

$$g'(z) - zg(z) = |z| - \sqrt{2/\pi}, \qquad |g| \le 1.$$

For $z \ge 0$ it is $g(z) = -1 + \sqrt{2/\pi}\, e^{z^2/2}\int_z^{\infty} e^{-v^2/2} dv$, extended oddly. Regression of the field Z on Z_r reduces the calculation to this one-dimensional identity. The regression derivative of A_{sT} is the covariance-weighted integral of h_0, so integration by parts gives

$$\mathbb{E}_V|Z_r| \le Cr^H + C\mathscr{D}\int_0^{\tau} \mathrm{Cov}(Z_r, Z_u)\mathbb{E}_V h_0(Y_{s+u}) du \le Cr^H + C\mathscr{D}r^{\alpha}x. \tag{98}$$

All derivatives are bounded at fixed L, τ because $A \geq m\tau$ and h_0 is bounded; finite Gaussian approximation justifies regression and integration by parts. The factor r^H comes from the untilted standard deviation. The last bound uses the refined covariance estimate (97).

Using $h_0(y) \geq \frac{1}{2}e^{-2|y|}$ and Jensen under $\mathbb{E}_V$ yields

$$x \geq a_0 \int_0^\tau r^{\alpha-1} e^{-C\mathscr{D}xr^\alpha} dr, \qquad a_0 = c_1 e^{-2K_s} > 0. \tag{99}$$

The bounded factor $e^{-C\tau^H}$ has been absorbed into c_1. Evaluating the integral gives

$$x^2 \geq \frac{a_0}{C\alpha\mathscr{D}}(1 - e^{-C\mathscr{D}x\tau^\alpha}). \tag{100}$$

If $C\mathscr{D}x\tau^\alpha \leq 1$, the inequality $1 - e^{-t} \geq t/2$ gives $x \geq ca_0\tau^\alpha$. Otherwise $x \geq c\sqrt{a_0/\mathscr{D}}$. Taking reciprocals in these two alternatives proves the first bound in (96). Since $\alpha < 1$, $\tau \leq T$, and (91) holds, the second follows from an envelope of the stated exponential-polynomial form. □

We next bound the variation of the call's exposure under the Black weights.

C.3. **Two conditional entropies and a finite-energy shift.** Fix $u \in [s, s+h]$, let $d = u - s$ and $v = T - u \geq \tau/2$, and condition on $\mathcal{G}_u$. The future field is

$$Y_{u+r} = \mu_u(r) + Z_r^u, \qquad 0 \leq r \leq v, \qquad |\mu_u(r)| \leq K_u =: K.$$

On this conditional future Gaussian space consider two separate densities

$$\nu_0 = \frac{V(L, A_{sT})}{\mathsf{N}_u}, \qquad \nu_1 = \frac{V(L, A_{sT})\Gamma_s}{\mathsf{M}_u}, \qquad H_j = H(\nu_j \mid \mathbb{Q}). \tag{101}$$

The prefix contribution to A_{sT} is fixed. Set

$$\ell = \mathscr{D}^{-1/(2\alpha)}, \qquad \gamma = \frac{2\alpha - 1}{2\alpha} > 0. \tag{102}$$

Lemma C.3. *If*

$$L^2 \geq M\tau + (M\tau)^2/4, \qquad d \leq \ell \leq v/2, \tag{103}$$

then

$$H_0 + H_1 \leq C(1 + K^2)\mathscr{D}^\gamma. \tag{104}$$

Proof. The first condition makes $V(L, A)$ increasing throughout $[m\tau, M\tau]$ by (81). Let

$$V_{\max} = V(L, A_{su} + Mv), \qquad \mathcal{U} = \log V_{\max} - \log V(L, A_{sT}) \geq 0.$$

Then

$$\mathcal{U} \leq C\mathscr{D} \int_u^T (M - f(Y_r)^2) dr. \tag{105}$$

On fresh Brownian time $[0, v]$ introduce the deterministic (given $\mathcal{G}_u$) Cameron–Martin drift

$$a_\ell(r) = A_0(1 + K)\ell^{-\alpha}(1 + r/\ell)^{-\alpha}[1 + \log(1 + r/\ell)], \tag{106}$$

where A_0 is a large fixed constant. The induced positive Volterra shift is

$$g_\ell(r) = c \int_0^r (r - z)^{\alpha-1} a_\ell(z) dz.$$

For $R = r/\ell \geq 1$, the integral over $z/\ell \in [R/4, R/2]$ gives a lower bound on the scaled convolution. There the power factors are at least $R^{\alpha-1}$ and $(3R/2)^{-\alpha}$, and $1 + \log(1 + z/\ell) \geq \frac{1}{4}[1 + \log(1 + R)]$. Consequently

$$g_\ell(\ell R) \geq \frac{c(2/3)^\alpha A_0}{16}(1 + K)[1 + \log(1 + R)]. \tag{107}$$

Choose A_0 so that $g_\ell(r) \geq K + 2\log(1 + r/\ell)$ for $r \geq \ell$. For $0 \leq r \leq \ell$ the same convolution is bounded above by $C(1 + K)$.

Let $\mathbb{Q}^a$ be this shifted conditional Gaussian law. Its entropy is

$$H(\mathbb{Q}^a \mid \mathbb{Q}) = \tfrac{1}{2} \int_0^v a_\ell(r)^2 dr \leq C(1 + K^2)\ell^{1-2\alpha}. \tag{108}$$

The profile has finite energy for every fixed $\alpha > 1/2$, because

$$\int_0^\infty (1+x)^{-2\alpha}[1+\log(1+x)]^2 dx = \frac{1}{b} + \frac{2}{b^2} + \frac{2}{b^3}, \qquad b = 2\alpha - 1 > 0. \tag{109}$$

The substitution $t = \log(1+x)$ reduces the integral to exponential moments. The entropy formula follows from the finite-dimensional Gaussian change of mean for step drifts and then L^2 approximation.

Under the shifted law the expected variance deficit is $O(\ell)$. The interval $[0, \ell]$ contributes at most $C\ell$. For $r \geq \ell$, (18), (107), and $\operatorname{Var} Z_r^u \leq CT^{2H}$ imply

$$\mathbb{E}_{\mathbb{Q}^a}[M - f(Y_{u+r})^2] \leq C(1 + r/\ell)^{-4}.$$

Indeed the predictor is at least $-K$, and the shift dominates it; the normal exponential moment contributes only a fixed constant. Integration and (105) show

$$\mathbb{E}_{\mathbb{Q}^a}\mathcal{U} \leq C\mathscr{D}\ell. \tag{110}$$

Since $\nu_0 = e^{-\mathcal{U}}/\mathbb{E}e^{-\mathcal{U}}$, $H_0 \leq -\log \mathbb{E}e^{-\mathcal{U}}$. The comparison inequality (67) with $\mathbb{Q}^a$ then gives

$$H_0 \leq C(1+K^2)\ell^{1-2\alpha} + C\mathscr{D}\ell \leq C(1+K^2)\mathscr{D}^\gamma. \tag{111}$$

For H_1, a separate lower estimate for Γ_s is essential. Since $d \leq \ell$ and the exponent is negative,

$$\begin{aligned} \Gamma_s &\geq c\int_0^\ell (d+r)^{\alpha-1} h_0(Y_{u+r}) dr \\ &\geq c_2 \ell^\alpha \exp\left(-\frac{2}{\ell}\int_0^\ell |Y_{u+r}| dr\right). \end{aligned} \tag{112}$$

Here $(d+r)^{\alpha-1} \geq (2\ell)^{\alpha-1}$ and Jensen supplies the last line. Under $\mathbb{Q}^a$, the average absolute field on $[0, \ell]$ has expectation at most $C(1+K)$, by the initial upper bound for g_ℓ and the bounded Gaussian variances. Thus

$$\mathbb{E}_{\mathbb{Q}^a} \log \Gamma_s \geq \alpha \log \ell - C(1+K). \tag{113}$$

For $F = e^{-\mathcal{U}}\Gamma_s$, we have $\log F \leq \log(C\tau^\alpha)$, since $\Gamma_s \leq C\tau^\alpha$. Applying (67) and combining (108), (110), and (113) gives

$$H_1 \leq C(1+K^2)\mathscr{D}^\gamma + \alpha \log(\tau/\ell) + C(1+K). \tag{114}$$

Since $\mathscr{D} \geq 1$ and $\tau \leq T$, the logarithmic term is bounded above by $C + C\log \mathscr{D} \leq C_\alpha \mathscr{D}^\gamma$. This proves (104) for both densities. □

The shift supplies lower bounds on conditional partition functions. Future pricing expectations remain under $\mathbb{Q}$. Finiteness of (109) prevents an extra mesh-dependent logarithm for fixed $H > 0$. Its divergence at $H = 0$ is also why this argument supplies no boundary-uniform estimate.

C.4. **Entropy smoothing and conditional derivatives.**

Lemma C.4. *There is a fixed sufficiently small $\eta > 0$ such that, on states satisfying*

$$h(1+L_s^2)(1+K_s^2)\tau^{-2} \leq \eta, \tag{115}$$

for $s \leq u \leq s+h$ and almost every $z \in [s,u]$,

$$|D_z \mathsf{N}_u| \leq C\frac{\sqrt{1+L_s^2}}{\tau}(1+K_u)\mathsf{N}_u, \tag{116}$$

$$|D_z \mathsf{M}_u| \leq C\frac{\sqrt{1+L_s^2}}{\tau}(1+K_u)\mathsf{M}_u + C\tau^{2H}\mathsf{N}_u. \tag{117}$$

Proof. Conditional derivative commutation from Appendix A gives

$$D_z\mathsf{N}_u = \mathbb{E}[V_A\Gamma_z \mid \mathcal{G}_u], \qquad D_z\mathsf{M}_u = \mathbb{E}[V_A\Gamma_z\Gamma_s + VD_z\Gamma_s \mid \mathcal{G}_u]. \tag{118}$$

All functionals belong to the derivative domain: L, τ are fixed, $A_{sT} \geq m\tau$, and (92) controls the singular-kernel derivatives. The last term in the second identity is bounded by $C\tau^{2H}\mathsf{N}_u$ and will remain additive.

There are three regimes. If $L^2 < M\tau + (M\tau)^2/4$, then $\mathscr{D} \leq C_T/\tau$. Since $\Gamma_z \leq C\tau^\alpha$, direct use of (118) gives coefficient $C\tau^{\alpha-1} \leq C_T/\tau$, sufficient for the assertions. If V is increasing but $\ell > v/2$, then $\mathscr{D}\tau^{2\alpha} \leq C$ because $v \geq \tau/2$. Hence $\mathscr{D}\Gamma_z \leq C\sqrt{\mathscr{D}}$, again sufficient.

In the remaining regime V is increasing and $\ell \leq v/2$. After reducing the constant η in (115) by a fixed factor if necessary, eligibility implies $h\mathscr{D} \leq 1$. Since $\mathscr{D} \geq 1$ and $1/(2\alpha) < 1$, it follows that $h \leq \ell$. Thus Lemma C.3 applies with $d \leq h \leq \ell$.

Let X now denote the fresh Brownian increment on $[u, u+\ell]$, a centered Gaussian of variance ℓ. Gaussian integration by parts gives the two distinct identities

$$\frac{\mathbb{E}_{\nu_0}X}{\ell} = \frac{1}{\ell}\int_u^{u+\ell} \mathbb{E}_{\nu_0}[(\log V)_A\Gamma_w]dw, \tag{119}$$

$$\frac{\mathbb{E}_{\nu_1}X}{\ell} = \frac{1}{\ell}\int_u^{u+\ell} \left\{\mathbb{E}_{\nu_1}[(\log V)_A\Gamma_w] + \frac{\mathbb{E}[VD_w\Gamma_s \mid \mathcal{G}_u]}{\mathsf{M}_u}\right\} dw. \tag{120}$$

These identities use the weighted derivative $VD_w\Gamma_s$, whose integrability follows from (92). By (93), replacing Γ_w with Γ_z costs at most $C\mathscr{D}\ell^\alpha$, since $|w - z| \leq 2\ell$. The additional source in (120) costs at most $C\tau^{2H}\mathsf{N}_u/\mathsf{M}_u$.

The entropy-normal estimate (68) and Lemma C.3 yield

$$\frac{|\mathbb{E}_{\nu_j}X|}{\ell} \leq \sqrt{2H_j/\ell} \leq C(1+K_u)\sqrt{\mathscr{D}}, \qquad \mathscr{D}\ell^\alpha = \sqrt{\mathscr{D}}. \tag{121}$$

Here $\gamma + 1/(2\alpha) = 1$. Substituting these estimates into (119)–(120), then into (118), and using (91) proves both assertions. In the M estimate the original prefix derivative of Γ_s adds another term of the same additive size, absorbed into the constant. Gaussian finite-coordinate approximations and closure justify all differentiations and integration-by-parts limits; no relative derivative bound for Γ_s is assumed. □

C.5. **Absorbing the random predictor before Poincaré.** Set

$$a_s^2 = C(1+L_s^2)(1+K_s^2)/\tau^2. \tag{122}$$

Lemma C.5. *On the eligible states* (115)*, after decreasing* η*,*

$$\mathbb{E}_s\mathsf{N}_u^2 \leq C\mathsf{N}_s^2, \qquad \mathrm{Var}_s\,\mathsf{N}_u \leq Cha_s^2\mathsf{N}_s^2, \tag{123}$$

$$\begin{aligned} \mathrm{Var}_s\,\mathsf{M}_u &\leq Cha_s^2\mathsf{M}_s^2 + Ch\tau^{4H}\mathsf{N}_s^2, \\ \mathbb{E}_s\mathsf{M}_u^2 &\leq C(\mathsf{M}_s^2 + h\tau^{4H}\mathsf{N}_s^2). \end{aligned} \tag{124}$$

Proof. Put $a_*^2 = C(1+L_s^2)/\tau^2$. Squaring and integrating the derivative estimates on $[s,u]$, with $K_u \leq K_s + A_{s,u}^{\mathrm{pred}}$, gives, for $X = \mathsf{N}_u$,

$$\mathbb{E}_s\|DX\|^2 \leq Cha_*^2\left\{(1+K_s^2)\mathbb{E}_sX^2 + \mathbb{E}_s[(A_{s,u}^{\mathrm{pred}})^2X^2]\right\}. \tag{125}$$

For $X = \mathsf{M}_u$ there is also $Ch\tau^{4H}\mathbb{E}_s\mathsf{N}_u^2$. Lemma C.1 bounds the predictor product. The resulting derivative term has coefficient $Cha_*^2h^{2H}$, which is below $1/2$ for small enough η: eligibility controls ha_*^2, and $h^{2H} \leq T^{2H}$. It can therefore be absorbed into the left-hand side of the derivative estimate. The remaining h^{2H} multiple of $\mathbb{E}_sX^2$ is absorbed into a fixed T-dependent constant times $1+K_s^2$. Consequently

$$\begin{aligned} \mathbb{E}_s\|D\mathsf{N}_u\|^2 &\leq Cha_s^2\mathbb{E}_s\mathsf{N}_u^2, \\ \mathbb{E}_s\|D\mathsf{M}_u\|^2 &\leq Cha_s^2\mathbb{E}_s\mathsf{M}_u^2 + Ch\tau^{4H}\mathbb{E}_s\mathsf{N}_u^2. \end{aligned} \tag{126}$$

Conditional Poincaré and the means $\mathbb{E}_s\mathsf{N}_u = \mathsf{N}_s$, $\mathbb{E}_s\mathsf{M}_u = \mathsf{M}_s$ now allow absorption of Cha_s^2 times the variances, again by choosing η small. The N inequality yields (123). Substituting that bound in the M inequality gives (124). At fixed L, τ the underlying functionals and their required derivatives have the domain bounds already proved, so the multiplier and Poincaré steps need no lower bound for their initial positive means. □

C.6. **Independent stock noise and the moving exposure.** On eligible states $\chi = Ch(1 + L_s^2)/\tau^2$ in (76) is small. Set

$$r_s = \mathsf{N}_s/\mathsf{M}_s = \mathbb{E}_s V/\rho_s \le \mathcal{K}_s/\tau. \tag{127}$$

Conditional variance decomposition using (74), the exact stock-mixture estimates (76)–(77), and Lemma C.5 gives

$$\mathbb{E}_s[(\mathsf{F}_u/b_s - 1)^2] \le C[ha_s^2 + h\tau^{4H} r_s^2], \qquad \mathbb{E}_s P_u^2/b_s^2 \le C r_s^2. \tag{128}$$

For the first inequality, total conditional variance splits into $S_s^2 \operatorname{Var}_s \mathsf{M}_u$ and a W variance bounded by $(e^\chi - 1)S_s^2 \mathbb{E}_s \mathsf{M}_u^2$. Since $\chi \le Cha_s^2$ and is small, this proves the stated bound. This calculation is valid for each positive mixture, with future option moments still under $\mathbb{Q}$.

The deterministic translation bound gives

$$|b_u - \mathsf{F}_u| \le Ch^\alpha P_u. \tag{129}$$

Therefore

$$\mathbb{E}_s[(b_u/b_s - 1)^2] \le C\{ha_s^2 + (h\tau^{4H} + h^{2\alpha})r_s^2\} \le C\mathcal{K}_s^2 h/\tau^2. \tag{130}$$

The last step uses $2\alpha > 1$ and bounded $h, \tau \le T$, with an enlarged exponential-polynomial envelope.

To control stock-delta leakage, we also need the weighted second moment

$$\mathbb{E}_s[(1 + L_u^2)b_u^2]/b_s^2 \le C(1 + L_s^2)\{1 + (h\tau^{4H} + h^{2\alpha})r_s^2\}. \tag{131}$$

Indeed the W mean of b_u is $S_s\mathbb{E}[V(L_s, A_{sT})\Gamma_u \mid \mathcal{G}_u]$, whose difference from $S_s\mathsf{M}_u$ is at most $Ch^\alpha S_s \mathsf{N}_u$. The weighted positive-mixture estimate (77), followed by (123)–(124) and the squared difference bound, proves (131). Here the change from Γ_s to Γ_u is controlled by the translation bound.

C.7. **Stock-delta leakage and the all-state cell bound.** The delta martingale formula and (88) remain valid for the non-Markov rough history by Appendix A. Conditional Itô isometry and Gronwall, followed by (128), (131), and $T - u \ge \tau/2$, give

$$\frac{\mathbb{E}_s[(Z_u^\delta)^2]}{b_s^2} \le Ch\left\{r_s^2 + \frac{1 + L_s^2}{\tau^2}[1 + (h\tau^{4H} + h^{2\alpha})r_s^2]\right\}. \tag{132}$$

Eligibility controls the apparently higher product:

$$\frac{1 + L_s^2}{\tau^2}(h\tau^{4H} + h^{2\alpha}) \le C(\tau^{4H} + h^{2\alpha - 1}) \le C_T. \tag{133}$$

Hence (127) reduces (132) to

$$\mathbb{E}_s[(Z_u^\delta)^2]/b_s^2 \le C\mathcal{K}_s^2 h/\tau^2. \tag{134}$$

The conditional competitor with call units $1/b_s$ and stock units $-\delta_s/b_s$ has Brownian coefficients b_u/b_s in the B direction and $\sigma_u Z_u^\delta/b_s$ in the W direction. Thus conditional minimization gives

$$h\zeta_{s,h} \le \mathbb{E}_s \int_s^{s+h} [(b_u/b_s - 1)^2 + \sigma_u^2 (Z_u^\delta/b_s)^2] du.$$

As in the nonnegative proof, common decision localization validates this quadratic comparison without imposing an unconditional inverse-exposure moment. The admitted global strategy is the projection (27).

Equations (130) and (134) imply $\zeta_{s,h} \le C\mathcal{K}_s^2 h/\tau^2$ on eligible states. On ineligible states, enlarging the fixed envelope so that its square dominates $(1+L_s^2)(1+K_s^2)/\eta$ allows the bound $\zeta \le 1$ to give the same estimate. We have proved on all states

$$\zeta_{s,h} \le \min(1, C\mathcal{K}_s^2 h/\tau^2), \qquad h \le \tau/2. \tag{135}$$

This proves the rough part of Lemma 5.1. The averaging argument in Lemma 5.2 uses contraction of the past Gaussian covariance and is uniform in target smoothing. Together these estimates provide the rough-case upper control used in Theorem 2.1.

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